

A course in language creation



Conlangs University

Phonology 4

Sound change



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Hello there, people of the Conlangs University! This week, we will be talking about language evolution with a special focus on sound changes and how sounds evolve over time.

Prerequisites

Given the nature of this lesson, it is highly recommended to have read all of the prior Phonology lessons - we will be focusing a lot on features and using shorthand notation that has been explained before.

Languages over time

Languages change over time! While this may seem slightly straightforward, you'd be surprised about how many people have never thought about it, or straight up refuse to believe it. For example, we have the Romance languages as a single group, all being descendants from Latin. We can thus say that Latin diverged and evolved into all these different languages. However, a language doesn't have to split into multiple descendants to evolve! For example, Modern English is significantly different from Old English, even if a certain degree of divergence can be seen in terms of dialects around the world. While Ancient Greek may still be approachable by Modern Greek speakers, the pronunciation has actually changed significantly. The Ancient Greeks used the onomatopoeia βῆ (bê) /bê:/ for the cry of a sheep - in Modern Greek, this would be read as /vi/! Doesn't sound like a sheep much, does it?

Sound changes

The way the sounds of a language evolve can be described in a roughly chronological order through what we call **sound changes**. Remember the phonological rules you saw before? Well, we're going to be using them a lot. There's different kinds of categories when it comes to these changes and we'll discuss all of them in detail with a bunch of examples later. However, let's first have a look at how these changes actually work!

Say we have a word /ap/ and we have the following change:

$p > f / _ \#$

As you probably remember from last lesson, this means that we'll end up with [af].

Not entirely sure why? I'd recommend going back and reading the last Phonology lesson again because this is a relatively simple environment. The main question right now is if this truly is a diachronic sound change or merely allophony. After all, we can perfectly predict when this occurs, right? We could totally just represent this word as <ap> /ap/ [af]. So what exactly turns this into a more interesting case?

This is where phonemic versus phonetic comes in. The moment we actually get a new distinction, we can say this sound change has **phonemicised**. Assume we start with two words, /apa/ [apa] and /ap/ [af]. Let's say we lose all word-final vowels:

$V > \emptyset / _ \#$

We now have an issue, because we would now be contrasting [ap] and [af]! This means that this change is now no longer predictable, so we can no longer consider it to be allophony. See how the timing of changes is relevant? If we would have lost the word-final vowels first, they would have merged into [af].

Keep in mind that many of the changes and categories we will be discussing below can be applied allophonically in your conlangs!

What else?

There are many, many other things that affect language evolution such as [analogy](#), [semantic shifts](#), [grammaticalisation](#) and so on. We'll talk about them in a very in-depth manner in a later lecture because it would result in a massive information overload. If you're curious, Wikipedia has some very nicely written articles on them, but unless you plan on making a naturalistic *a posteriori* language, they probably won't be as relevant for you.

Assimilation & Dissimilation

The core concept of assimilation and dissimilation is pretty straightforward - sounds either becoming more similar or more different to another sound. Assimilation is significantly more common, but we'll find plenty of examples of dissimilation as well. We can distinguish progressive and regressive examples of both changes: if a feature spreads backwards, we consider it to be regressive - if it spreads forwards, we call it progressive.. Regressive is significantly more common, but not the only option.

Assimilation

Let's try to pronounce the English word <infinity>. We would probably expect the nasal to be realised as [n] based on the orthography, right? Pay close attention and you'll notice that your lips are actually involved. However, they don't actually entirely close - what kind of sorcery are we dealing with here?

The answer is that our nasal sound /n/ is assimilating in POA with the following consonant: the labiodental [f]. The result of this regressive assimilation is the labiodental nasal [ɱ]. Nasal assimilation in this fashion is near-universal and you will find it in almost all languages. Other examples in English include <stink> with [-ŋk] - however, this nasal is also represented in broad transcription, i.e. /-ŋk/, because of the phonemic status of /ŋ/ (e.g. <sing> /sɪŋ/).

Regressive assimilation

When Vulgar Latin evolved into Italian, /p k/ followed by /t/ assimilated into them, leading to geminates. This change can be notated as follows:

$p\ k > t_t$

Examples include Latin *octo* > Italian *otto* (eight) and Latin *subtus* > Italian *sotto* (bed). Why the in *subtus* is affected? Because it's purely orthographical and the actual cluster is /pt/.

We can also find these kind of changes over a distance. For example, Italo-Celtic languages had the following change:

$p...k^w > k^w...k^w$

This explains why Latin has *quinque* < Proto-Indo-European **pénk^we*. The Germanic languages, on the other hand, irregularly had progressive assimilation in this word only, leading to Proto-Germanic **fimfe* where **f* is the natural outcome of Proto-Indo-European **p*. Note how the notation of the nasal changes to match the labial aspect of the following consonant, even if the actual pronunciation would have been [ɱf]. Likewise, Latin would have had [ŋk^w] for the corresponding cluster.

Assimilation is not only restricted to consonants - the Germanic umlaut is a very good example of this. If you know some German, you'll know that the plural of *Mann* /man/ is *Männer* /mene/. While the actual productivity and distribution in German is slightly more complicated, it's a good example of Germanic i-umlaut. A fossilised but probably more familiar example in English is how the plural of *man* is *men* and how one *foot* become two *feet*. How, you ask? Let's have a look!

In Proto-Germanic, we have the words **mann* and **fōts* for man and foot, respectively. They look similar enough, right? Their plural forms were **manniz* and **fōtiz*. The next step is Germanic i-umlaut, which dictates (at least in English) that these back vowels fronten if followed by **i* or **j*. More concretely, we end up with a stem distinction, resulting in sg. *mann* but pl. *mænni* and sg. *fōt* but pl. *fōti*. Predictable enough, right? Unfortunately, this suffix was later lost, leading to these sounds becoming phonemic. As such, the only distinction between singular *mann* and plural *mænn* was the vowel. These pairs regularly evolved into *man/men* and *foot/feet*.

Progressive assimilation

As mentioned earlier, progressive variants are significantly less common. An example of adjacent progressive assimilation is Proto-Indo-European **-ln-* assimilating to **-ll-* in both Germanic and Italic (the regressive counterpart would be **-nn-*). An example is PIE **k_lnis* "hill" becoming Latin *collis* and Proto-Germanic **hulliz*. In Germanic, we can also see the application of i-umlaut outside of the plural forms: **hulliz* > *hylli* > English *hill*.

Vowels may also once again partake in progressive assimilation. However, this is far less common than the regressive umlaut-like changes discussed above, and often a part of **vowel harmony** (of which umlaut is an example) - a far more complex process that will not be discussed in another lecture.

Coalescence

Coalescence, also known as fusion, is perhaps one of my favourite sound changes. What happens is that two adjacent sounds merge into a single sound, combining distinctive features from both. This is particularly common for vowels, where a diphthong transforms into a long vowel positioned around the middle ground between the diphthong components. Examples include [ae] > [e:] (Latin *aethēr* "ether" > Spanish *éter*) and [au] > [o:] (Latin *aurum* "gold" > Spanish *oro*) - the second can still be seen orthographically in eg. English *law* [lɔ:], even if the exact positioning of the result on the vowel chart is slightly different.

For more funky changes, this may also happen to consonants! For example, Latin *bis* (twice) comes from an original *dwis*, where the *dw* has fused, inheriting the voiced plosive aspect from /d/ and the labial aspect from /w/, resulting in /b/. Proto-Indo-European also experienced **g^w* becoming /b/ in Greek - an example is βούς (*boûs*), where the first part is actually responsible for the bu- in butter! Oscan (an Italic language and as such a sister language) had this too, and when Latin borrowed the word for cow from them, they ended up with *bōs*. Knowing that the oblique stem of this word was actually *bōw-*, we can also see that this eventually ended up as French *bœuf* before appearing in English as *beef*. Fun fact - this is actually a doublet of cow!

The existence of retroflex consonants may also often be ascribed to coalescence: for example, Norwegian <rt> /ɾ/ still very clearly shows the original cluster, in which the rhotic has introduced the retroflex feature to the voiceless stop, which was previously dental.

Nasalisation

Nasalisation can appear through numerous means. A first is clearly coalescence of a vowel and a nasal consonant, in which the consonant passes its nasality onto the vowel and disappears. A second would be assimilation, where a vowel becomes nasalised when adjacent to a nasal consonant, without the need for that consonant to be elided. Either way, nasalisation of vowels most commonly occurs because of assimilation - the other possibility, rhinoglottophilia, is discussed later on.

Dissimilation

The concept of dissimilation is quite clear: as the opposite of assimilation, it makes sounds that are similar more different. Note that this does not mean that a sound is simply yeeted somewhere else in the vowel or consonant charts - rather, a similar distinctive feature is replaced while others remain the same. The fact dissimilation is significantly less common is clear in how it's harder to find examples, and is generally almost exclusively diachronic in nature.

Regressive dissimilation

Greek and Indo-Iranian underwent Grassman's law, which is a simple sound change that dictates that when two adjacent syllables both start with an aspirated consonant, the first one is deaspirated. This is especially visible in the perfect tense, where reduplication is used heavily and Grassman's law is still relevant. For example, φύω (phúō) /pʰý.o:/, which means "I grow", has the perfect πέφυκα (péphuka) /pépʰyka/ - you can see that this originally would have been *phéphuka. This sort of regressive dissimilation at a distance is by far the most common form of dissimilation.

We may find an example of the rare regressive dissimilation in adjacent segments in German *sechs* /zeks/. German <ch> is generally used for the fricative /x/, which must have dissimilated from the following fricative /s/.

Progressive dissimilation

Progressive dissimilation at a distance is reasonably common. We can find a bunch of examples in English words that are originally loans - for example, *purple* can be traced back to Latin *purpura*, and shows that the second rhotic has dissimilated into a lateral. You may find that you spell, pronounce or both *cardamom* as if it were *cardamon*. English *marble* also originally comes from Latin *marmor*, showing not only a dissimilation of the second rhotic, but also of the second nasal. Additionally, Russian has февраль /fevralʲ/ for February from an older Old Church Slavonic февроуари (*fevruari*).

We may find rare examples of progressive dissimilation in continuous segments in the evolution from Latin to Spanish. For example, the Latin accusatives *hominem* and *nomen* (man and name, respectively) ended up as Spanish *hombre* and *nombre*. Those seem to be radically different and you're probably scratching your head over it. It's true that there's more involved than simple dissimilation here, so we're keeping the actual answer to this puzzle for later - however, for the sake of keeping this a good example, we'll be seeing a cluster /mn/ dissimilating into /mr/.

Unpacking

Unpacking is essentially the opposite of coalescence - splitting a sound into two, dividing features across them. This very regularly happens when a word with a nasal vowel is borrowed into a language that doesn't have them - examples are plentiful, but include English *monsoon* from Portuguese *monção* /mõsẽũ/. If the nasal component would not have been realised as a nasal consonant, it would have been lost.

If a syllabic consonant ejects a vowel, this may be considered a form of unpacking as well. For example, Sanskrit ऋ [ɾ̥] regularly became [ɾɪ] in Hindi, where the rhoticity is expressed by the rhotic but the syllabic aspect by the vowel.

Vowel breaking is a specific subtype of unpacking where a (often long) vowel splits into a diphthong or even a triphthong. This is the reason why in English, we pronounce the letter *i* as /aɪ/ (from a historic /i:/) - this also explains why despite sounding so differently, *is* and *ice* aren't so dissimilar after all if you know the final -e in *ice* is nowadays used to indicate length, as opposed to the short vowel in *is*. Spanish very typically has /je/ and /we/ for stressed *e *o - Latin *petra* (stone), *focus* (fire) > Spanish *piedra*, *fuego*.

Epenthesis & Elision

Epenthesis and elision refer to the addition and deletion of sounds, respectively. While there's a bunch of terminology for specific instances of epenthesis and elision, it's actually relatively easy to grasp!

Epenthesis

There's a fancy way to split epenthesis into two groups, being excrescence for when a consonant is added and anaptyxis when a vowel is added. Additionally, when a sound is added at the beginning of a word, we may call it prothesis, while paragoge is the equivalent term for sounds being added at the end of a word. As much as we're cool kids, we're not going to be bothering with these terms overly much because it's pretty obvious which kind of epenthesis we're dealing with.

Basically, adding vowels tends to be for a pretty straightforward reason, which is to simplify pronunciation, often altering words to make them accord to a language's phonotactics. Note that what is considered to be legal phonotactically may change and cause epenthesis to take place!

Excrescence

This is where we can start explaining part of our Spanish *hombre y nombre* examples. We previously said that an original /mn/ cluster would have dissimilated into /mr/. Try saying [amra] 10 times as quickly as you can. Are you sure you didn't accidentally say [ambra] at least once? In this case, epenthesis is used to ease the pronunciation by bridging consonant clusters. There are many more examples of this, such as English *hamster* occasionally being pronounced as if it were *hampster*, and *warmth* being pronounced as if it were *warmpth*.

We may also find this in the Greek word *ambrosia* - you know, the food of the gods that makes you immortal. Greek has *a-* as equivalent to English *in-* and *un-*, i.e. as a negative prefix. Knowing that the *-b-* is epenthetic, we can work from a form *mrosia*. If we ditch the *-sia* for now, we end up with *mro-*. Seems suspiciously similar to Latin *mori* (to die), doesn't it? That's in fact correct, and we can analyse Greek *ambrosia* as immortality from the adjective *ambrotos* (immortal).

We can have more fun by looking at Proto-Germanic **þunraz* (thunder). I'm sure you very quickly figured out that the *-d-* in English *thunder* is epenthetic, as is the case for Dutch *donder*. German lacks this and has *Donner* instead - Old Norse instead decided to roll with regressive assimilation, leading to *þórr* - which also happens to be deified, and that's Thor for you!

The English phenomenon of linking and intrusive R is also an example of epenthesis to ease pronunciation, in this case by bridging a hiatus. If you're a Beatles fan, you're sure to recognise the song "A Day in the Life" from their 1967 Sgt. Pepper's Lonely Hearts Club Band album. In one of the verses, they

very clearly sing "I saw-r-a film". As much as the orthography points towards a consonant that already bridges any possible hiatus, the actual pronunciation of *saw* is /sɔ:/.

Anaptyxis

We can divide anaptyxis into two main sections, one where it affects native words and one where it affects loans.

Native words

Many languages tend to lose unstressed word-final vowels at one point or another (this will be talked about more elaborately in elisions), and this would often result into illegal word-final clusters. These often end up being resolved through the introduction of a so called **prop vowel**, which is generally very neutral and often the schwa /ə/. For example, compare Vulgar Latin *acre* (sharp) and *carru* (cart) giving French *aigre* and *char*, respectively. Often this prop vowel is not inserted at the end but slightly before, especially in the case of sonorants, cf. Proto-Germanic **akraz* giving English *acre* - while the vowel is written at the end, we all know the pronunciation is [-ər]. Speaking from personal experience, I'm not sure how widespread this is, but in many Flemish dialects, a word such as *warm* (warm) is actually bisyllabic with a schwa being inserted between the rhotic and the nasal. If a suffix is attached, e.g. *warne*, the need for insertion is no longer necessary.

English also has synchronic insertion for plurals and past forms, compare *glass* and *glasses* to *bow* and *bows*, or *walk* and *walked* to *want* and *wanted*. In the plurals, the insertion is represented orthographically, while for the past forms, there is a often silent vowel.

At the end of Latin as a single unity, an epenthetic /i/ was inserted at the beginning of words that started with #sC, which surfaces as *e-* in all Romance languages. For example, you have Latin *schola* (school) > Spanish *escuela*, French *école* (where *é-* represents a historic *es-*).

Loans

Japanese is often considered a good example of using prop vowels to break up illegal foreign clusters. Examples include キャップ /kjaɸɸ/ (from English *cap*) and ストリート /suto:li:to/ (from English *street*). Brazilian Portuguese has /meki'dõnawdʒis/ for McDonalds, where the vocalisation (see later) of /l/ to /w/ and the /dʒ/ for <d> before /i/ are regular.

Elision

Elision is also sometimes called deletion. Like epenthesis, there are dedicated terms for where in the word elision takes place: *aphaeresis* takes place in the beginning, *syncope* in the middle and *apocope* at the end.

The loss of vowels happens quite regularly and may not always be reflected orthographically. Examples include English *family* /fæmli/. French very often elides unstressed vowels, most commonly schwas, and uses apostrophes to represent said elided vowels: *ce est* > *c'est* and *nous ne avons pas* > *nous n'avons pas*.

Historically speaking, syncope explains the final puzzling question regarding Latin *hominem* > Spanish *hombre*, being the loss of the unstressed *-i-*, which allowed the cluster /mn/ to exist and further dissimilate.

A variant of elision is haplogy, which is described as the elimination of one of two similar, adjacent syllables. For this reason, it is often humourously referred to as haplogy. This explains things such as *gently* > *gently*, *Englaland* > *England* and the colloquial pronunciation of *library* as /laɪbrɪ/. It's possible to consider haplogy to be a form of dissimilation as well.

Cluster reduction is also an important application of elision. If one would end up with illegal clusters, either because of historical reasons or because of loans, one could patch them up with prop vowels through elision, or reduce them through elision. For example, English has reduced word-final */mb/ and */ŋg/ to /m/ and /ŋ/; compare e.g. *lamb* and *sing*. AAVE often reduces clusters to single stops at the end of words, leading to *friend* to rhyme with *Ben* and *coal* to be homophonous with *cold*.

Often, sounds may be elided for no real reason at all - for example, French lost a vast majority of its word-final consonants and often vowels too, e.g. *aiment* /ɛm/. Proto-Oceanic lost most of its word-final consonants, resulting them to be only present when a transitive suffix is added, as they were not word-final in this case. An example would be Proto-Oceanic **inum* (drink), which becomes *unu* in Boumaa Fijian (but *unuma* with the transitive suffix) and *inu* in Longgu (but *inuvia* with the suffix).

Important here is that the loss of sounds may lead to compensatory lengthening. As the name implies, this means that when a sound is deleted, a **preceding** (but not following) sound may be lengthened to retain the prosody. Note the focus on may, as this is not strictly necessary. This explains why English *Vgh* very often marks a historically long vowel: *night* is pronounced the same way a hypothetical *nite* would be (where the unpronounced final *-e* marks length instead). Non-rhotic dialects tend to actually lengthen most vowels when the rhotic is lost. French is a good example here, as VsC environments were affected. English *forest* is a cognate with French *forêt*, where the loss of *s* lengthened the preceding vowel. Despite phonemic vowel length being lost in most of the French dialects, it is still notated orthographically and may be realised as a different vowel quality instead.

Lenition & Fortition

Lenition and fortition are two categories of sound changes that almost exclusively affect consonants. Lenition refers to sounds becoming more sonorous, while fortition refers to the opposite. Like before, one of the two is significantly more common than the other - in this case, lenition is the more common one.

Lenition

Lenition can be divided into two subcategories, depending on through which means a sound lenites: opening and sonorisation. While these can represent the scale of a sound from very sonorous to not sonorous at all, in reality, lenition tends to happen in step and switch back and forth between these scales.

Opening

In this type of lenition, the name "opening" actually refers to the articulation. At the least sonorous side, we find geminate stops - solely by losing their gemination, they become more sonorous. An example would be Latin *cattus* (cat) > Spanish *gato*.

The next step in opening up is affrication, in which a voiceless stop becomes an affricate. Some sounds have multiple options: for example, [t] may either become [t͡s] or [t͡θ]. German shows this quite clearly, with words such as *Pferd* /pfe:rt/ next to Dutch *paard* /pa:rt/.

The next step is spirantisation (also known as deaffrication), in which only the non-plosive element of the affricate remains, i.e. [t͡s] > [s]. When coming from another source, affricates such as [t͡ʃ] may also spirantise to [ʃ] - this explains how Spanish has [t͡ʃ] for <ch> while Portuguese and French have [ʃ]. It also explains how German *lass* is cognate to English *let*.

The penultimate step is debuccalisation, in which the sound is reduced to a placeless approximant, generally glottal in nature. This is evident in Greek initial aspiration being a remnant of Proto-Indo-European *s-: compare Greek ἑπομαι (hépomai) with Latin sequor (the change of *k^w to *p is another nice example of coalescence). Word-final /s/ is realised as [h] in various varieties of Latin American Spanish. In various English dialects, /t/ may be reduced to [ʔ] in various environments - this, too, is an example of debuccalisation, even if the affrication step has been skipped.

Once debuccalisation has taken place, there is generally speaking no way back (except through rhinoglottophilia, which we'll talk about later), which means that the next step is total elision. The very common tendency to lose [h] is an example of this.

Old Japanese /p/ has mostly shifted to Japanese /h/ - however, it is realised as [ɸ] before /u/, where the bilabial feature is a remnant of the original plosive. /h/ later also underwent a change where it became [ç] before /i j/, leading to three possible realisations of /h/. This loss of /p/ however did not occur in

geminate, leading to the existence of doublets regarding 日本 (Japan), leading to the possible pronunciations of /nihon/ (from /p/) and /nipˈpon/ (from /pp/).

Sonorisation

Sonorisation is more closely linked to voicing, as voiced sounds are always more sonorous than their voiceless counterparts. This type of lenition happens particularly often intervocalically (between vowels), in which sense the adoption of voicing could be seen as assimilation to the surrounding vowels. This type of lenition was grammaticalised across word boundaries in Celtic languages, leading to alterations such as Scottish Gaelic *cas* /kʰas/ "steep" > *glé chas* /kleː xas/ "very steep".

As previously implied, the first step is the voicing of voiceless consonants. This happened in most of the Romance languages, leading to pairs such as Latin *vita* (life) > Spanish *vida* and Latin *lupa* (female wolf) > Spanish *loba*.

The second step is approximation, in which voiced stops first become continuants before becoming full on approximants. Once again, sounds may exhibit different outcomes: [b] may become either [β] or [v] (although this may be attributed to the relative instability of [β] as a phoneme) and [d] may become [ð] or [z] (cf. above) or even [r] as a tap. This tap can be seen in the execution of the dentals in the common pronunciation of English *butter* in Canadian and American English. Full-on approximation can be seen in [ɣ] surfacing as semivowels such as [u j w]. The only thing more sonorous than these approximants/semivowels would be elision.

A particularly common case of sonorisation is l-vocalisation, in which [l] vocalises into [i] or [w]. This is a productive rule in Portuguese, where the plural of *animal* is *animais*. It also happened historically in Dutch, explaining how Dutch *oud* and English *old* are cognates.

Fortition

Less commonly known as strengthening, fortition does the opposite of lenition - in other words, it follows the same steps, but in a reverse order. For example, fricatives becoming stops or voiced stops becoming voiceless count as fortition. The cross-linguistically rare dental fricatives may also be substituted for plain stops, as has happened in all Germanic languages except for English and Icelandic, although several English dialects include this substitution, which is referred to as th-stopping. Word-final devoicing, which is reasonably common cross-linguistically, is also an example of fortition.

Very rarely, you may find vowels undergoing fortition to actually become consonants. For example, Miyakoan had /i/ (from a historical /i u/ merger) which became /ɟ/ (or ɟ after /b g/ through voicing assimilation). This leads to very curious cognates between Miyakoan and Japanese, such as /pʂtu/ (person) from **pito* (Japanese ひと /hito/), /kʂmu/ (liver) from **kimo* (Japanese きも /kimo/) and /tabʂ/ (journey) from **tabi* (Japanese たび /tabi/).

Occasionally, you may find that a very sonorous sound just jumps back all the way to one of the least sonorous sounds. This is occasionally analysed as being an assimilation of the consonantal feature,

although not everyone may agree with this analysis. An example of this obviously rare change are some central Romansh dialects, where the following change occurred:

$j \text{ w } ə > k / V_$

This process remains mostly productive, leading to alterations such as /kreja/ (believes) ~ /krekr/ (to believe). Cypriot Greek experienced a similar change where /j/ becomes [c] or [k] when following consonants: /aðerfi/ "brother" (Standard Greek αδελφός *adelfós* /aðelfɔs/), when receiving the plural suffix /-a/, becomes /aðerfka/ (through an intermediate /-fja/).

Palatalisation

Palatalisation refers to any change that palatalises a consonant and is in many cases conditioned by a front vowel or [j] - as such, this counts as assimilation. However, the exact functionality tends to be more in line with lenition, leading to some to consider this a hybrid form. In some cases, palatalisation also refers to vowels fronting - this has previously been covered in regressive assimilation.

A very familiar example in English is the pronunciation of *did you* as [dɪdʒu:] rather than [dɪdju:], which is called yod-coalescence in English linguistics. Historically, English velar stops were palatalised into /tʃ dʒ/ before front vowels - an example would be *church* (which was *ċirice* in Old English), which corresponds to Dutch *kerk* and German *Kirche*. This is also the reason you may sometimes find alterations such as *speak* and *speech* or *cold* and *chill*. Additionally, several unpalatalised doublets from Northumbrian and Old Norse have been borrowed into English after the shift, such as *kirk* being a Northern English and Scottish dialectal variant of *church*, or the alteration between *ditch* and *dike*.

The Romance languages experienced an even larger amount of palatalisation, which didn't just affect the velar stops, but also dental stops and even nasals (e.g. Latin *sĭmius* "monkey" > French *singe* /sɛ̃ʒ/). Gallo-Romance languages, of which French is the most well known, also included palatalisation of velar stops before [a]: Latin *carus* (dear) > French *cher* /ʃɛʁ/.

In some rare cases, palatalisation may occur without any triggering front sound. For example, in Western Romance languages, the cluster [kt] was palatalised through vocalisation to [jt] (Latin *octō* "eight", *noctem* "night" > French *huit* /ɥi/, *nuit* /nɥi/, Portuguese *oito*, *noite*), after which a regular conditioned palatalisation changed the cluster into [tʃ] for Spanish *ocho*, *noche* and into [tɕ] for Romansh *otg*, *notg*. Additionally, Latin clusters involving a plosive and [l] experienced unconditioned palatalisation, leading to oddities such as Latin *clāmāre* (to call) > Spanish *llamar* (with initial /ʎ/), Italian *chiamare* (with initial /kj/) and Portuguese *chamar* (with initial /ʃ/).

Other changes

The following categories are other common sound changes or tendencies that do not deserve as much attention as assimilation, dissimilation, lenition and fortition. However, this definitely does not mean they're not as common or important!

Metathesis

Metathesis refers to two sounds being swapped within a word. For example, English *foliage* is often pronounced as if it were *foilage*. Slightly less common is metathesis over distance, also called hyperthesis, which is apparent in the common but irregular interchange between /r ~ l/ from Latin to Spanish: *parabola* > *palabra*, *periculum* > *peligro*. English also showcases this relatively irregularly, leading to *asterisk* being pronounced as *asterix* and *cavalry* as *calvary*. *Bird* and *horse* were originally *bridd* and *hross* while *wasp* was *waps*. French has *moustique*, meaning mosquito - the order English shows here is authentic.

French has the informal speech pattern called *verlan*, in which entire syllables are deliberately swapped. The name itself is the *verlan* equivalent of *l'envers*, meaning "the reverse", and it features words such as *chelou* from *louche*. Lunfardo, an argot of Spanish in Buenos Aires, has a Spanish equivalent called *vesre* - like French, from an original *revés*.

Historically, Ancient Greek featured metathesis of semivowels when following sonorants: φαίνω (phainō) "I appear" < /p^hanjo:/. The original form is still visible in the aorist ἐφάνη (ephānē) "he appeared". The Slavic languages are especially noteworthy for their liquid metathesis, such as Serbo-Croatian *grād* "castle, town" as cognate to English *yard* or Serbo-Croatian *rāme* "shoulder" as cognate to English *arm*.

A special application of metathesis is quantative metathesis, in which not sounds are exchanged, but quantity or vowel length. An example of this would be Ancient Greek *ἦο (*ēo*) > εῶ (*eō*). Less commonly, this may also cause interactions between vowels and consonants: compare Latin *littera* (whence English *letter*) to *lītera* (whence English *alliteration* - the initial *al-* is an assimilation of the Latin prefix *ad-*).

Rhotacism

Rhotacism is a relatively rare change that turns a consonant, generally a voiced alveolar one, into a rhotic sound. This most commonly affects [z], explaining not only why Latin word such as *flōs* (flower) have an oblique stem *flōr-* (with the [s] having been voiced to [z] as part of intervocalic lenition before experiencing rhotacism), but also the alteration between *I was* and *we were* in English, as well as the correspondence between Latin *-us* and Icelandic *-ur*. Portuguese also had significant rhotacism affecting /l/, explaining *branco* (white), *obrigado* (thank you, cognate with English *obliged*) and *prato* (plate). Neapolitan has rhotacism affecting intervocalic or initial /d/, leading to pairs such as Latin *dentem* (tooth) > Neapolitan [rɛndə].

Rhinoglottophilia

While we have mentioned nasalisation before, a far less common but interesting approach to nasalisation is through rhinoglottophilia. This term, which is not related to rhinoceroses, describes the connection between the laryngeal (glottal) and nasal articulations. This sounds pretty cryptic, but let's look at how the Kennedys pronounce Harvard: [hã:vəd] See how the vowel next to /h/ is nasalised? This is rhinoglottophilia in a nutshell. It also explains why some languages may have the following slightly odd shift:

h > ŋ

Additionally, a few languages such as Cantonese exhibit the epenthesis of /ŋ/ word-initially when a word used to start with a vowel - this too can be explained as rhinoglottophilia, where the velar nasal replaces a previously allophonic glottal stop.

What about the vowels?

Besides umlaut, the reference to vowel harmony, coalescence and vowel breaking, we haven't really looked at how vowels change all that much. The main reason for this is that they generally tend to do what they'd like to do and the rules aren't quite as strict as for consonants. For example, Dutch had a historic shift where *u* became /y/, despite there not immediately being any sort of pressure. Like with consonants, how balanced an inventory is may affect vowels - if you only have five vowels that are pretty close to each other on the chart, they may spread apart a bit to increase the distinction. Vowels may occasionally just merge without a problem - the merger of /y/ into /i/ is attested in many, many languages with English, Icelandic and Greek being examples.

Vowels may adapt to their environment - for example, nasal vowels may have the tendency to centralise (compare the French oral vowels /a i u o/ to their nasal counterparts /ã ĕ œ õ/) or they may backen when next to liquids.

So what now?

First things first, this list is not exhaustive. As you may have realised, sounds can go to many, many places. A question many beginners ask is whether sound A can become sound B, to which the answer is always yes, **given enough time**. For example, the Armenian cognate of English *two* is երկու (erku)! How, you may ask? We've seen all the components so far, so let's have a look at a late Proto-Indo-European form, which would be /dwo:/. [o] and [u] are reasonably close enough on the vowel chart, and length can be lost without traces, so we're mainly concerned with how [dw] managed to become [erk]. The rhotacism of [d] to [r] is a good first step, and matches what we've seen so far. The introduction of initial [e] is an example of prothesis, which is a subform of anaptyxis, which is a subform of epenthesis. All that's left now is the fortition of [w] to [k] for us to reach our final destination. If you didn't believe that sound changes on their own could be wild, now you should!

If you'd like to start piecing together your very own list of sound changes, try to first and foremost keep the scope of time in mind. Languages can be extremely conservative (compare Icelandic and Old Norse), extremely innovative (French does a pretty good job at not sounding like Latin at all - the only way the connection with Latin remains clear is through its incredibly conservative and archaic orthography) and anything in between. If you're outdoing French, however, you may want to reconsider a few things. I have done my best to roughly label each change with a degree of rarity - once again, there are no single truths. Is fortition rare? Sure, it is, but that doesn't mean you should avoid it - after all, one of the key features of Germanic languages as a unity is a massive systematic fortition that separates it from the other Indo-European branches. It's merely a reference to use in an attempt to keep your proportions in check.

Perhaps the most important thing to keep in mind is that changes tend to happen in a systematic matter. Having only one sound affected all the time is bound to make things look very artificial and unnatural. For example, if you want to voice intervocalic [p] to [b], why not also voice [t] to [d]? It's things like this that will make everything feel much more coherent.

Secondarily is the aspect of how balanced phoneme inventories are. This is a rather complicated matter that is not exactly my duty to explain in this lecture, but unstable sounds will be likely to shift to unused, more common phonemes. This may not necessarily fall under any of the categories discussed above (how would you classify [d > v]?) but that doesn't mean they're not relevant at all. The same goes for vowels. Keep in mind that while your inventory keeps shifting around, so does its balance, so make sure you're on the lookout for imbalances - they may very well fuel your inspiration!

Index Diachronica

A resource that always pops up at some point is the [Index Diachronica](#). By including it in this lecture, I hope to be able to get rid of some very popular misconceptions right from the beginning.

The Index Diachronica aims to be a compilation of all documented sound changes of all languages. This, combined with a nifty interface that allows you to select a sound and see into what it may evolve or where it may come from, means it's an incredibly exciting tool that's sure to blow your mind when you use it for the first time. However, not all is perfect, as this site comes with a few very major drawbacks.

First of all, the nature of being a compiled source renders the site/database as a secondary resource - this means that academically, like Wikipedia, it should not be used as a source. Sure, you may think, but I use Wikipedia for all my linguistics lookups and I'm just fine! The issue is that the Index Diachronica majorly suffers from sourcing issues - not all sources may be copied diligently and in many cases, you'd be lucky if the source was mentioned or the link was functional at all. Additionally, it incorporates suggested sound changes for a whole bunch of hypothetical or disproven families, meaning that some of the things you may take as truth are flawed guesswork at best. As such, it is important not to take things you find on the site as truth, but to solely use it as a device to fuel your creativity - if you'd like to check if a change has indeed happened, either make sure the source is academic or try to find an academic source yourself.

Homework!

We're more than 6500 words later, which means it's time to try to summarise what you've learnt in this document. There's been an awful lot of technical terms so feel free to look up and check if you forgot something. The following homework features some general exercises to help you memorise how everything works, as well as some exercises specific to your conlang.

1. The following changes are attested in real languages. Try to describe them as well as you can:

Example: $n > m / _C[+labial]$ (/n/ undergoes regressive assimilation of labial features)

$p \ t \ k > f \ \theta \ x$

$e > a / _C[+resonant]a$

$sj \ zj > \int \ ʒ$

$rs > s:$

2. Slightly more difficult: try to analyse which sound changes have happened in the following examples (the order may be important!) - sound change notation plus the category is sufficient:

Example: $mu:si > mi:s$ ($my:si$ [assimilation] $> my:s$ [apocope] $> mi:s$ [unrounding])

$kida > giða$

$statjō > staʃon$

$nikto > ipi:to$

$namno > navon$

3. Have a look at the allophones you decided for on your conlang and in which environments they appear. Can you classify them under any of the categories above?
4. Take ten random words from your conlang and try to make ten random sound changes before applying them. What do you see? Are all words affected equally much? Does it happen to merge grammatically distinct forms?